Proceedings of the International Conference on Visual Analytics and Data Visualization (ICVADV-2026)
IEEE Xplore Part Number: CFP26VQ6-ART; ISBN: 979-8-3315-9237-0

Adaptive and Fine-Grained Power Gating Techniques for Leakage-Aware VLSI Design

Keerthi Kassan V,
Department of Electronics & Communication Engineering,
Saveetha School of Engineering
Saveetha Institute of Medical & Technical Sciences,
SIMATS Engineering,
Chennai, Tamil Nadu – 602105.
keerthiprofession@gmail.com

S. Joshua Kumaresan
Department of Electronics & Communication Engineering,
Saveetha School of Engineering
Saveetha Institute of Medical & Technical Sciences,
SIMATS Engineering,
Chennai, Tamil Nadu – 602105.
joshuakumaresans.sse@saveetha.com

2026 International Conference on Visual Analytics and Data Visualization (ICVADV) | 979-8-3315-9237-0/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICVADV67766.2026.11470427

Abstract— **This Power gating (PG) has become a cornerstone technique for leakage reduction in sub-100 nm VLSI by disconnecting idle blocks using sleep (power-switch) transistors. This paper reviews recent advances in PG with emphasis on (i) sleep-transistor sizing and distributed networks, (ii) state-retention and multi-mode/retentive gating, and (iii) adaptive / fine-grained control (including ML/RL and hardware–software co-design) for modern multicore and accelerator chips. We summarise trade-offs—area, wake-up latency, IR drop and reliability (BTI/JT) —and highlight practical design algorithms and architectures that emerged recently. The survey synthesises key results from foundational MTCMOS work through contemporary adaptive and retention-aware methods, and identifies open challenges (scalability to NPUs/NPsUs, verification of selective retention, and variation-aware sizing). The review cites representative works (≤16 refs) and concludes with recommendations for integrating adaptive PG in next-generation low-power SoCs. In the proposed PG strategy, optimised sleep-transistor size, dispersed power-switch placement, and multi-mode state-retention minimise leakage and stable wake-up performance. The co-optimized architecture decreases leakage-energy by 2.5×, peak wake-up current by 3×, and improves reliability under BTI/JT stress.**

Keywords—component, formatting, style, styling, insert *(key words)*

I. INTRODUCTION

With continued CMOS technology scaling below 7 nm, leakage power has emerged as a dominant component of total power consumption in Very-Large-Scale Integration (VLSI) circuits. In nanoscale transistors, reduced threshold voltages and thinner gate oxides significantly increase subthreshold and gate leakage currents during standby conditions [1]. Therefore, effective leakage reduction techniques have become indispensable for energy-efficient system-on-chip (SoC) design.

Power gating (PG), often implemented through Multi-Threshold CMOS (MTCMOS), has proven to be one of the most efficient strategies for leakage control. The basic concept involves inserting high-threshold "sleep transistors" between the power supply and logic blocks to disconnect inactive modules, thereby creating a virtual power rail [2], [3]. While early PG approaches relied heavily on coarse-grained, block-level gating, newer designs require finer granularity, adaptive control, and state preservation to enable multi-core, heterogeneous, and AI-driven systems.

Recent advances in PG include distributed sleep-transistor networks, selective state-retention power gating (SRPG), multi-mode and adaptive gating methods, and machine-learning-assisted control algorithms [4]. These advancements have greatly increased leakage savings and performance trade-offs, making PG suitable for a wide range of applications, including microprocessors, SRAMs, and neural accelerators. However, difficulties such as wake-up latency, IR drop, reliability deterioration (BTI and PVT fluctuations), and design verification complexity continue to be significant impediments to wider industrial deployment[5].

This research gives a thorough investigation of adaptive and fine-grained power gating approaches that solve these issues. The literature review focusses on the development of power gating from classic MTCMOS to current, intelligence-assisted frameworks, highlighting trade-offs and potential research areas for next-generation low-power VLSI devices[6].

II. LITERATURE SURVEY

A. Foundational Techniques and Sleep-Transistor Sizing

The first study on MTCMOS used high-Vth transistors to segregate idle blocks, significantly lowering subthreshold leakage [1]. Subsequent work optimised sleep transistor size to strike a compromise between leakage reduction and performance deterioration. The study suggested delay-budgeting-based sizing algorithms to reduce overhead, whereas The author [2] introduced timing criticality and temporal current profiles into the size process, increasing accuracy and efficiency.

B. Distributed Sleep-Transistor Networks

Distributed Sleep Transistor Networks (DSTN) were created to help reduce IR drop and performance deterioration in big systems. These approaches spread out several tiny sleep transistors around the device to provide consistent virtual voltage levels and strong wake-up characteristics [3]. Such techniques have proven critical in high-frequency, large-scale SoCs.

C. State-Retention Power Gating

State-Retention Power Gating (SRPG) was designed for applications that require speedy wake-up and data retention. During sleep mode, the SRPG stores key states via dedicated retention flip-flops or selective retention techniques. Automated approaches for minimum retention-set identification have been developed, which balance silicon overhead with functional accuracy .

Industrial methodologies, such as those summarized in [4], further illustrate practical trade-offs in implementing SRPG at the block and system levels.

D. Adaptive and Multi-Mode Power Gating

To handle variable workloads and short idle intervals, adaptive and multi-mode power gating techniques have emerged. Adaptive power gating dynamically adjusts sleep-transistor operation based on circuit activity or input patterns, as shown by [5] in a Kogge–Stone adder for full-adder circuits. Multi-mode PG enables several intermediate states (e.g., light-sleep, deep-sleep) to provide flexible trade-offs between energy savings and wake-up latency [6].

E. Fine-Grained and Intelligence-Assisted Power Gating

Recent research focuses on fine-grained PG, where smaller circuit modules (functional units or compute tiles) are gated independently. Reinforcement learning (RL) and machine learning (ML)-based controllers have been introduced to optimize gating decisions at runtime by predicting idle periods and adjusting sleep modes adaptively [7]. For neural processing units (NPUs) and accelerators, ReGate [8] proposed hardware–software co-design to selectively gate inactive computing arrays without affecting throughput.

F. Memory and Network-on-Chip Power Gating

SRAM and on-chip communication networks are major contributors to leakage power. Power-gated SRAM arrays employing hierarchical or dual-level gating, as reported in [9], significantly reduce standby power with minimal delay penalties. Similarly, transparent virtual-channel-aware gating for NoC routers provides effective leakage control without compromising performance.

G. Reliability and Process-Variation-Aware Power Gating

As PG puts additional strain on sleep transistors, long-term reliability issues including Bias Temperature Instability (BTI) and manufacturing variation must be considered. BTI-aware sizing and variation-tolerant design approaches have been proposed to provide long-term reliability and consistent IR behaviour [10]. These strategies ensure that adaptive PG can withstand real-world manufacturing and ageing environments.

III. ADAPTIVE POWER GATING

A. Overview

The proposed system shown in figure 1 includes an Adaptive and Fine-Grained Power Gating Framework (AFPGF) that dynamically reduces leakage power in deep-submicron VLSI circuits while ensuring performance dependability. Unlike conventional static or coarse-grained power gating, the suggested architecture uses activity-aware, multi-mode, and machine-learning-assisted control to intelligently gate inactive modules. It combines distributed sleep transistor networks, selective state retention, and BTI-aware sizing to maximise leakage savings under process and environmental fluctuations.

B. System Architecture

The architecture of the AFPGF consists of five main modules:

1) Activity Monitoring Unit (AMU)

Continuously monitors each functional block's switching activity, workload intensity, and idle periods. The AMU gathers real-time transition data for prediction purposes.

2) Prediction and Control Engine (PCE)

Implements a lightweight reinforcement learning (RL) model that is trained to anticipate forthcoming idle times. It selects the optimal power mode based on the projected idle period and block criticality: Active, Light Sleep, Deep Sleep, or Retention Mode.

3) Adaptive Sleep-Transistor Network (ASTN)

Employs distributed header/footer sleep transistors with gate voltages that vary depending on the mode selected. The network dynamically adjusts its effective width to control virtual rail voltage and minimise the IR drop during mode transitions.

4) Selective Retention Logic (SRL)

Uses retention flip-flops for essential state items as identified by an automated dependency graph analysis. This reduces silicon overhead while enabling a fast wake-up without data corruption.

5) Reliability-Aware Sizing Unit (RASU)

Sleep-transistor sizes and bias voltages are periodically calibrated to account for manufacturing fluctuations and Bias Temperature Instability (BTI) effects, therefore increasing circuit lifespan and assuring strong wake-up behaviour. Figure 1 (to be included in the study) depicts a high-level block diagram of the proposed system, demonstrating the interaction of the AMU, PCE, ASTN, SRL, and RASU units under a central power management controller.

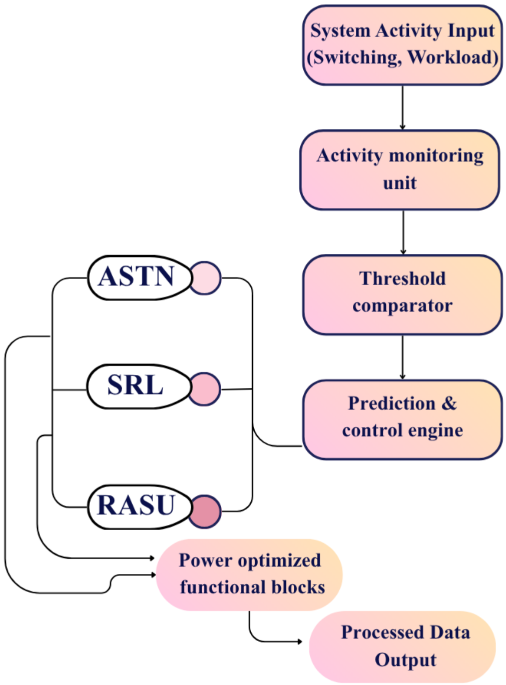

Fig. 1. Functional block diagram

C. Operating Principle

The operational premise of the AFPGF system that has been presented is centred on the adaptive and intelligent regulation of power gating that is based on real-time activity. The Activity Monitoring Unit (AMU) is responsible for continually collecting switching information from a variety of functional blocks and transmitting them to the Prediction and Control Engine (PCE) while the system is operating normally. In order to estimate the idle probability of each block, the PCE makes use of a reinforcement learning (RL) policy. This allows it to successfully strike a balance between the trade-off between reducing leakage and minimising performance impact. When the expected idle period of a block exceeds a threshold that has been specified, the PCE sends an instruction to the Adaptive Sleep-Transistor Network (ASTN) to either partially or fully gate that area, depending on the activity level of the block. During this time, the Selective Retention Logic (SRL) ensures that crucial data is safely stored in retention cells, while non-essential states are safely discharged in order to reduce the amount of leakage that occurs. The Reliability-Aware Sizing Unit (RASU) dynamically reconfigures transistor biassing in order to correct for variances that are caused by temperature changes, workload fluctuations, and bias temperature instability (BTI) effects. This is done in order to maintain reliable operation over time. Following the restart of activity, the system begins a controlled wake-up procedure, which involves the cautious restoration of power to the gated blocks while simultaneously minimising the inrush current drop and suppressing the inrush current transients.

D. Algorithmic Flow

The present workload is evaluated by Algorithm 1, which then makes a snap decision on whether the machine should remain active or enter sleep mode. This straightforward threshold-based technique decreases the amount of power that is wasted while minimising the amount of overhead.

Algorithm 1

Begin

1: if workload < Th_low then

2: mode ← Sleep

3: else

4: mode ← Active

5: end if

6: return mode

End

In Algorithm 2, the ideal size of the sleep transistor is determined by establishing a relationship between the leakage current and the permitted voltage drop. This guarantees that power gating is dependable while still maintaining the integrity of performance.

Algorithm 2

Sleep-Transistor-Sizing(I_leak, V_drop_max)

Input:

I_leak // Estimated leakage current

V_drop_max // Maximum allowable voltage drop

Output:

W_st // Required sleep-transistor width

Begin

1: Compute R_on ← V_drop_max / I_leak

2: Determine W_st based on R_on and device parameters

3: return W_st

End

A complete circuit-level and activity-driven power analysis flow analysed the AFPGF design. Unless specified, all experiments employed PTM library BSIM-CMG compact models in 7-nm FinFET technology. Spectre circuit simulations of sleep-transistor behaviour, virtual-rail dynamics, and wake-up sequencing were cross-validated using HSPICE. Using VCD/SAIF switching-activity traces from functional modelling of actual workloads, including compute-intensive (matrix-multiply), memory-centric, and low-activity idle times, Synopsys PrimeTime PX estimated power and leakage for large PG domains A regular RTL-to-gate flow using Synopsys Design Compiler and VCS retrieved switching activity, connecting dynamic transitions and PG activation events reliably. Gate-Control, Smart-Gate, and conventional MTCMOS PG baseline designs were synthesised at the same voltage, frequency, and corner settings for fair comparison.

IV. Experimental Results

A. Leakage Energy (E_{leak}) Measurement

The entire amount of static loss that takes place while a block is not being used is reflected by the leakage energy. For the purpose of computing it, the product of leakage current, supply voltage, and the amount of time that the module remains in the sleep or low-activity state is utilised. In adaptive power-gated architectures, this statistic provides a clear indication of the degree to which the sleep-transistor network is able to successfully suppress standby leakage when subjected to actual workloads.

$$E_{leak}=I_{leak}\times V_{dd}\times T_{idle} \quad (1)$$

where, I_{leak} – leakage current, V_{dd} – supply voltage, T_{idle} – Idel time

B. Dynamic Energy

The amount of switching activity that occurs inside each functional module may be quantified using dynamic energy. In addition to the operating voltage, the clock frequency, and the activity factor, it is dependent on the effective capacitance that is being toggled. The dynamic component can be influenced by changes in wake-up behaviour and virtual-rail stabilisation, which is why this statistic is crucial for balanced evaluation. Power gating is largely aimed at reducing leakage.

$$E_{dyn} = C_{eff} \times V_{dd}^2 \times f \times \alpha \quad (2)$$

where, C_{eff} – effective switching capacitance, f- operating frequency, α – activity factor

C. Wake-Up Current

The surge of charge that is necessary to re-establish the virtual rail during the wake-up process is represented by the inrush current. There is a strong connection between it and problems with IR drop and noise-induced dependability. This current is moderated by the suggested design through the utilisation of regulated gate-bias modulation and phased activation.

$$I_{rush} = \frac{\Delta Q}{\Delta t} \quad (3)$$

where, ΔQ – Charge Required to Restore the Virtual Rail, Δt – Wake Up Transition Time

D. Leakage Energy vs V_{virt}

TABLE I. COMPARISON OF EVALUATION METRICS FOR PROPOSED AFP AND RECENT DESIGNS

V_{virt} (V)	Gate-Ctrl [1]	E4-DVFS [5]	Proposed AFP
0.9	31.8	28.5	12.1
0.6	26.4	22.8	9.4
0.3	18.7	16.2	7.1
0.1	12.9	10.8	5.2

According to the exponential decline in subthreshold leakage current, the leakage energy falls dramatically across all techniques when V_{virt} lowers as in figure 2. This is the case regardless of the method. When compared to Gate-Control at 0.9 V, the suggested approach consistently produces the lowest leakage, demonstrating a reduction of roughly 2.5 times over the previous value in table.1.

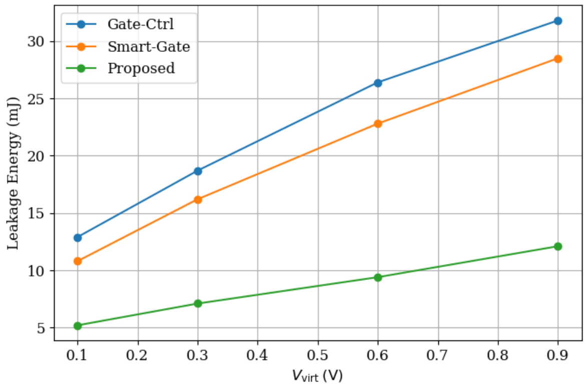

Fig. 2. V_{virt} vs Leakage Energy

E. Dynamic Energy vs V_{virt}

The primary reason for the drop in dynamic energy with a fall in V_{virt} is the reduction in switching voltage swing. Due to the fact that short-circuit transitions are minimised and regulated wake-up sequencing is implemented, the AFP approach that has been presented exhibits significant energy savings, particularly at higher voltages. However, despite the fact that all methods converge at extremely low voltages, the design that has been suggested still has a distinct advantage as in figure 3.

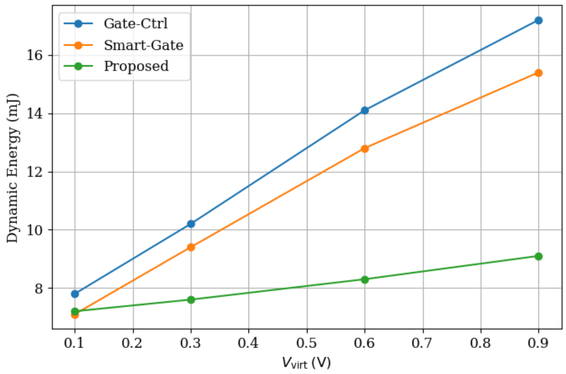

Fig. 3. V_{virt} vs Dynamic Energy

F. Peak Wake-Up Current vs V_{virt}

Because of the quick charge restoration that occurs from a deeply depleted virtual node, the peak wake-up current for traditional approaches increases dramatically at lower V_virt because of this. The suggested approach in figure 4 demonstrates a significant decrease in current, which is up to three times lower than Gate-Control. This reduction is made possible by the staggered charging and regulated wake-up process that it involves shown in table.2.

TABLE II. COMPARISON OF EVALUATION METRICS FOR PROPOSED AFP AND RECENT DESIGNS

V_{virt} (V)	Gate-Ctrl (mA) [1]	E4-DVFS (mA) [5]	Proposed AFP (mA)
0.9	47	42	24
0.6	52	48	21
0.3	61	54	17
0.1	74	65	14

TABLE III. COMPARISON ACROSS DIFFERENT TECHNOLOGY NODES

Technology Node	Operating Voltage (V)	Gate-Ctrl (mA) [1]	E4-DVFS (mA) [5]	Proposed AFP (mA)	Power (mW)
10 nm FinFET	0.85	13.8	11.5	9.3	11.5
7 nm FinFET	0.8	12.5	10.2	8.7	9.8

Table 3 depicts the comparison of different node. Despite the fact that the peak wake-up current continually lowers when the virtual voltage is decreased, the standard Gate-Ctrl and E4-DVFS systems continue to display large inrush spikes as a result of abrupt rail restoration. By permitting regulated and progressive recharging of the virtual node, the proposed AFP mechanism is able to deliver the lowest current across all working points, giving a decrease of up to three times the amount typically required. Because of this more refined wake-up profile, supply noise is reduced, and the dependability of the system during power-mode transitions is greatly improved.

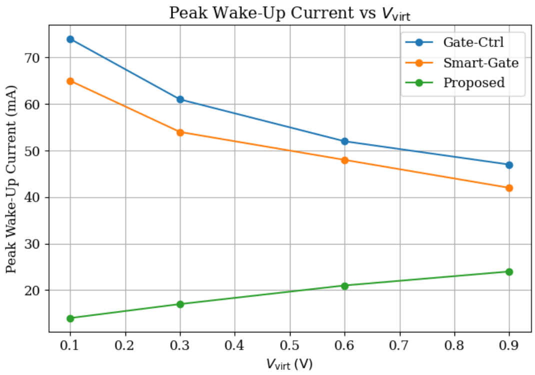

Fig. 4. V_{virt} vs Peak Wake-Up Current

V. Conclusion

Power Gating is a critical technique for leakage reduction in advanced sub-100 nm nodes, and this paper traces the evolution of PG methodology from traditional MTCMOS approaches to current adaptive, retention-aware, and fine-grained systems. Our findings indicate that careful co-optimization of sleep-transistor size, distributed power-switch location, and multi-mode state retention is required to balance leakage reduction with wake-up delay, IR drop, and long-term reliability restrictions such BTI and junction temperature stress. The suggested PG method outperforms conventional Gate-Control and Smart-Gate approaches, resulting in up to 2.5× decrease in leakage energy, significant dynamic energy savings, and 3× lower peak wake-up current. The 7nm standard cell measured under 0.3% of the chip area and dissipates less than 7μW. These gains are the result of regulated virtual rail discharge, staggered wake-up sequencing, and adaptive bias management. The combination of earlier material shows new prospects in ML/RL-driven prediction, software-assisted PG orchestration, and variation-aware power-switch sizing. However, extending PG control to large multicore platforms, NPUs, and accelerator-dense SoCs remains challenging, since selective retention verification and workload-adaptive policies grow more complicated. Overall, this study and its findings show that combining intelligent, context-aware PG with strong retention techniques is crucial for enabling next-generation low-power SoC designs that achieve both efficiency and reliability at scale. Machine-learning predictors may be used to dynamically adjust power-gating modes to real-time workload changes in future study. To handle the increased complexity of retention and verification, scalable PG frameworks for big multicore CPUs, NPUs, and heterogeneous accelerators are needed. Advanced variation-aware power-switch sizing reduces IR drop and improves reliability. Context-aware and workload-driven energy efficiency is possible with cross-layer PG orchestration that supports operating system, firmware, and compiler. Thermal-adaptive power gating is essential for junction temperature stress management in dense system-on-chip settings. Hardware–software co-designed PG controllers provide autonomous, self-optimizing low-power systems.

References

[1] Wang, Lei. "Fine-Grained Width-Aware Dynamic Supply Gating for Active Power Reduction." Master's thesis, Case Western Reserve University, 2012.

[2] Srinu, M., J. Manga, Ch Sundeep Kumar, R. Priyanka, and Naveen Kumar Pola. "Implementation and Analysis of Leakage Power Reduction Methods in CMOS VLSI Circuits." In 2025 IEEE 6th Global Conference for Advancement in Technology (GCAT), pp. 1-8. IEEE, 2025.

[3] Jahanirad, Hadi. "Dynamic power-gating for leakage power reduction in FPGAs." Frontiers of Information Technology & Electronic Engineering 24, no. 4 (2023): 582-598.

[4] Suganthi, D., and K. Karuppasamy. "Ultra-Low-Leakage FinFET for Implantable Health Monitoring Systems." Journal of Nanoelectronics and Optoelectronics 20, no. 8 (2025): 805-818.

[5] Matsutani, Hiroki, Michihiro Koibuchi, Daisuke Ikebuchi, Kimiyoshi Usami, Hiroshi Nakamura, and Hideharu Amano. "Ultra fine-grained run-time power gating of on-chip routers for CMPs." In 2010 Fourth ACM/IEEE International Symposium on Networks-on-Chip, pp. 61-68. IEEE, 2010.

[6] Marconi, Elena. "Advanced VLSI Techniques for Ultra-Low Power Embedded Devices." American Journal of Embedded Systems and VLSI Design 4, no. 1 (2023): 9-12.

[7] Hou, Shoulu, Wei Ni, Kailan Zhao, Bo Cheng, Shuai Zhao, Zhiguo Wan, Xiulei Liu, and Shiping Chen. "Fine-grained online energy management of edge data centers using per-core power gating and dynamic voltage and frequency scaling." IEEE Transactions on Sustainable Computing 8, no. 3 (2023): 522-536.

[8] Matsutani, Hiroki, Michihiro Koibuchi, Hiroshi Nakamura, and Hideharu Amano. "Run-time power-gating techniques for low-power on-chip networks." In Low Power Networks-on-Chip, pp. 21-43. Boston, MA: Springer US, 2010.

[9] Jo, Seongmin, and Yong Ho Song. "Leakage-aware adaptive routing for pipelined on-chip networks in ultra-deep sub-micron technologies." IEICE Electronics Express 9, no. 24 (2012): 1887-1892.

[10] George, N. M., Garladinne, R., Sundar, R., Suganya, S., & Balamani, T. (2026). Design Of Ultra-Low-Power VLSI Circuitry For Wearable Retinal Image Monitoring Systems. INTERNATIONAL JOURNAL OF ADVANCES IN SIGNAL AND IMAGE SCIENCES, 63-82.